

“There Is Always A **TRADE-OFF** Between **QUALITY** And **COST**”

Robert John Webber

field application engineer, Harwin



Electromagnetic (EMC) shielding prevents damage to electronic components by protecting PCB elements and associated hardware from electromagnetic interference (EMI). Robert John Webber, field application engineer, Harwin, discusses the role of interconnect devices and EMC shielding in manufacturing in an interaction with Ayushee Sharma

Q. What should one look for while choosing a connector?

A. You can choose any connector you want but there is always a trade-off between quality and cost. The first thing is to understand the problem. The component needs to be in line with what the customer is trying to do. If it is something that is all about getting it as cheaply as possible, prefer disposable ones. Where price is less of an issue, like military, focus should be more on getting the absolute best quality as it needs to be really robust.

The next step is to think about all the environmental issues around it, like temperature based on whether it will be used in a place as hot as a desert or a cold place like Siberia. Or, maybe it is a cheap device used in consumer electronics, and it will be used inside an office.

For example, connectors may be designed for signal or power contacts. The customer might have chosen connectors with bigger pins, which are designed for power; but if only low signals are being used and there is no power in it, then signal contacts could be used instead.

Q. What are the specific requirements for electric vehicles (EVs)?

A. EVs need to be robust. Higher range of connectors can be employed as EVs are not too cost-sensitive. So, safe, space-saving and lightweight components can be quite important

to an EV manufacturer, but these also need to have high power handling capacity. Hence, connectors should be such that these solve two different problems at the same time.

Q. What are the requirements for critical applications like aerospace and defence?

A. One of the biggest issues in this industry, whether it is military, aviation, marine or hybrid, is that every engineer wants customised solutions in addition to a very good fit. To allow clients to fulfil their needs without paying the extra price, they can be provided with a custom-made product in a standard product offering. This allows them to choose the type of combination they want.

Q. How is innovation in surface mount technology (SMT) leading to better PCB hardware components?

A. Using SMT means that we can free up space; and because we want things to be more miniaturised, it helps with that. It improves efficiency as well as overall component selection. Using SMT interconnect also saves cost of the manufacturing process. Though, some industries still prefer through-hole technology.

From a manufacturing point of view, we need to ensure the product conforms to necessary specifications and that it can be installed quickly. We have a department in our company that just looks at proofs-of-

concepts—understanding the market, seeing trends in the market, different problems that engineers from different industries face and so on.

Q. How is board-level shielding (BLS) different from traditional shielding solutions?

A. BLS is done to prevent EMI. Designers do not have to conform to the predefined geometry of the shield with BLS. BLS, when done without the usual secondary hand-soldering operation, can save time and expenses, and prevent damage.

Q. Can you discuss your offerings?

A. We have two types of product. On one hand, we have high-reliability-type connectors that are used in things like Formula One, aerospace, space technology, military, medical products.

On the other hand, we have small and cheap PCB components that are useful for solving design problems like putting things together in a small amount of space that includes EMC shielding.

Q. What is your roadmap for the future?

A. In general, our primary focus is on peripheral applications within Industry 4.0, apart from cloud computing, like artificial intelligence (AI), Big Data, robotics, the Internet of Things (IoT). The entire Asia (China, India, Singapore) is important for us. **EFY**